

1. Motivation & Research Gap

As AI-generated content becomes increasingly common in everyday digital environments, questions arise about how exposure to it affects users' critical engagement. Lee et al. (2025) examines when and how knowledge workers engage in critical thinking while using generative AI tools, and finds that higher confidence in AI is associated with reduced critical thinking effort. However, the study focuses on critical thinking *during active AI use* like evaluating outputs, refining prompts, verifying responses. It does not ask a prior question: whether encountering AI-generated content in the first place affects a user's tendency to question whether what they are reading is AI-generated at all. This suspicion about content origin is a separate psychological process from critical thinking during use, and it has not been directly examined.

A related study by Milička et al. (2025) finds that participants without corrective feedback made the most errors in AI text detection precisely when they felt most confident. However, the study treats detection as a trainable skill improved through feedback, it does not ask whether passive exposure alone, without any training or feedback, is sufficient to shift how suspicious people are about content they encounter. This leaves open the question of what happens in everyday life, where people encounter AI-generated content continuously and without any corrective signal.

Our work contributes to this framework by examining content exposure as an isolated variable specifically, whether encountering AI-generated content is itself sufficient to affect a user's subsequent decision to verify new content

2. Research Question & Hypothesis

Our research question is: Does a single exposure to AI-generated content reduce young adult's tendency to verify subsequent content, suggesting that familiarity with AI-generated content diminishes critical vigilance over time?

We hypothesize that participants who read AI-generated content in Round 1 will be less likely to use the AI detector in Round 2 compared to participants who read human-generated content in Round 1. This would suggest that even minimal exposure to AI-generated content is enough to begin eroding the impulse to scrutinize new content, a process that, at scale and over time, could have significant implications for how people engage critically with information they encounter online.

3. Variables & Operationalization

Table 1: Variables Overview

Variable	Description	Type	Measurement
Content Type (IV)	Type of text shown in Round 1	Between-subjects manipulation	Group 0 = human Group 1 = AI
Detector Use (DV)	Whether participant uses AI detector in Round 2	Behavioural measure Round 2 only	Binary: Yes / No
AI Tool Usage (Confound)	Frequency of AI tool use in daily life	Pre-questionnaire self-report	5-point scale (1 = never, 5 = daily)
Online Reading (Confound)	Habit of questioning content read online	Pre-questionnaire self-report	4-point scale (1 = never, 4 = always)
Self-Trust in AI Detection (Confound)	Participant's confidence in their own ability to detect AI-generated content	Operationalized via Round 1 manipulation — confirmed or undermined	Binary: raised / lowered. Recorded automatically via Pavlovian

Confounds are measured via pre-questionnaire or recorded automatically and controlled for statistically.

The independent variable (IV) is the type of text participants are exposed to in Round 1 - either human-written or AI-generated. This is manipulated directly by the experiment: Group 0 receives a human-written text and Group 1 receives an AI-generated text. Everything else about Round 1 is identical for both groups.

The dependent variable (DV) is the participant's verification decision in Round 2 specifically, whether they choose to use the AI detector when presented with a new AI-generated text. This is captured as a binary response: yes or no. Round 2 is identical for all participants, meaning any difference in detector use between groups can be attributed to what they were exposed to in Round 1.

We identify three confounds that could independently influence the DV and therefore need to be measured and controlled for statistically.

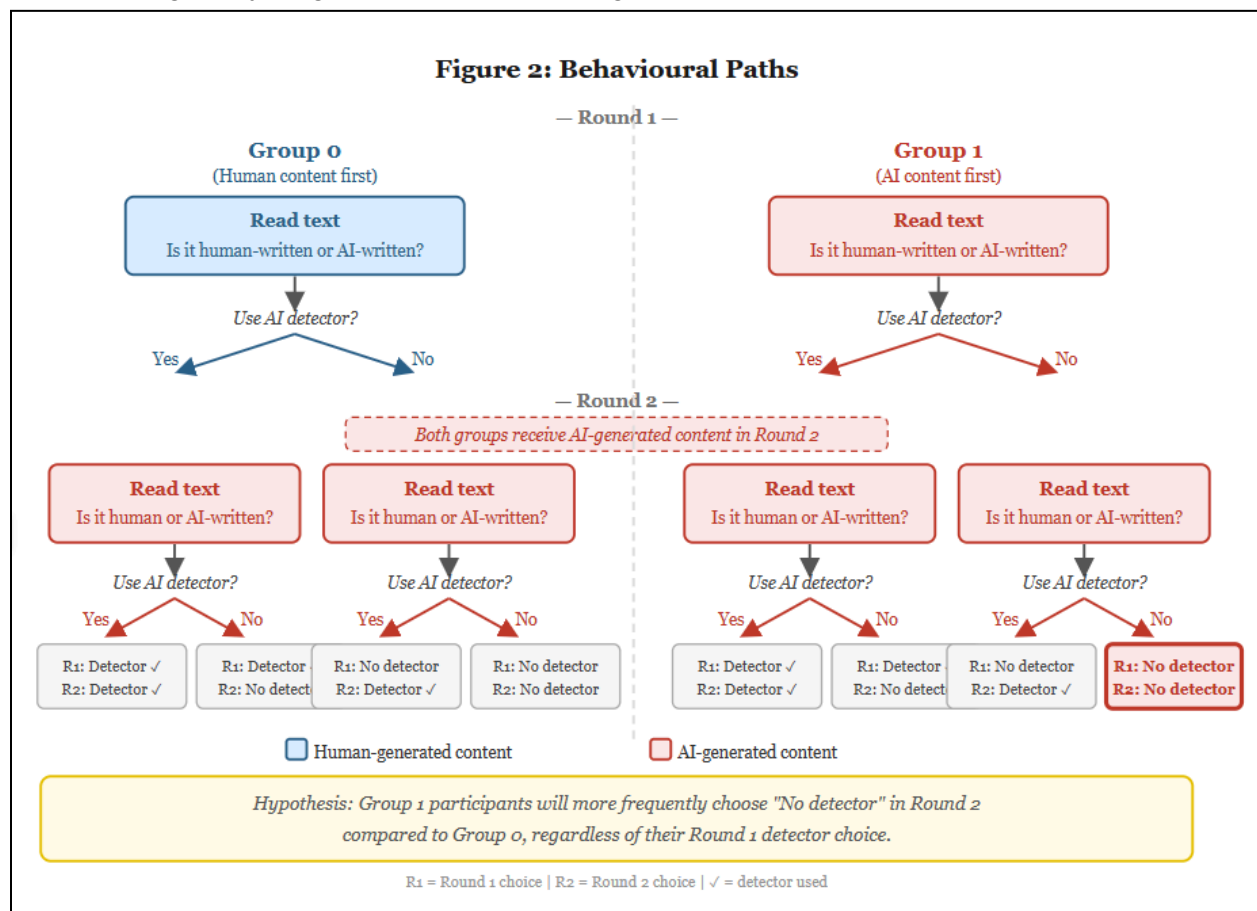
The first is daily AI tool usage. Participants who already use AI tools frequently in their everyday life may enter the study with a pre-existing reduced tendency to scrutinize AI-generated content, regardless of what they read in Round 1. This is measured in the pre-questionnaire on a five-point scale: 1 ("I never use AI tools"), 2 ("rarely"), 3 ("occasionally"), 4 ("often"), 5 ("daily or multiple times a day").

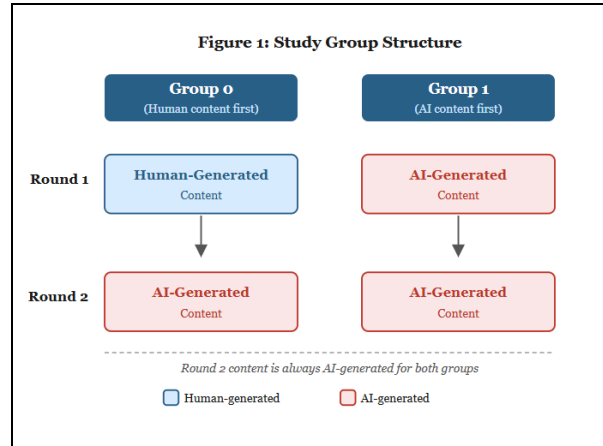
The second is online reading behaviour. Participants who habitually question what they read online are likely to use the detector regardless of group assignment, while participants who passively consume content may not use it regardless. This is measured in the pre-questionnaire on a four-point scale: 1 ("I never question what I read online"), 2 ("I sometimes do"), 3 ("I often do"), 4 ("I always question what I read online").

The third is the participant's trust in their own ability to detect/suspect AI-generated content. A participant who enters Round 2 feeling confident in their detection ability may behave differently from one who feels uncertain, regardless of what content type they saw in Round 1. This self-trust is operationalized through the Round 1 manipulation participants who chose to use the detector receive fabricated feedback confirming their judgment, raising their confidence in their ability to detect AI content, while participants who did not use the detector are told their judgment was wrong, lowering that confidence.

4. Study Design

The study is experimental, allowing us to examine whether content type in Round 1 directly influences verification behaviour in Round 2. Participants are randomly assigned to one of two between-subjects conditions at the start of the experiment: the human-content condition or the AI-content condition through Pavlovia's automated counterbalancing. Both groups go through the same two rounds, with Round 1 being the only point of difference between them and Round 2 being identical for all participants. This design isolates content type as the variable of interest while keeping everything else constant across groups.

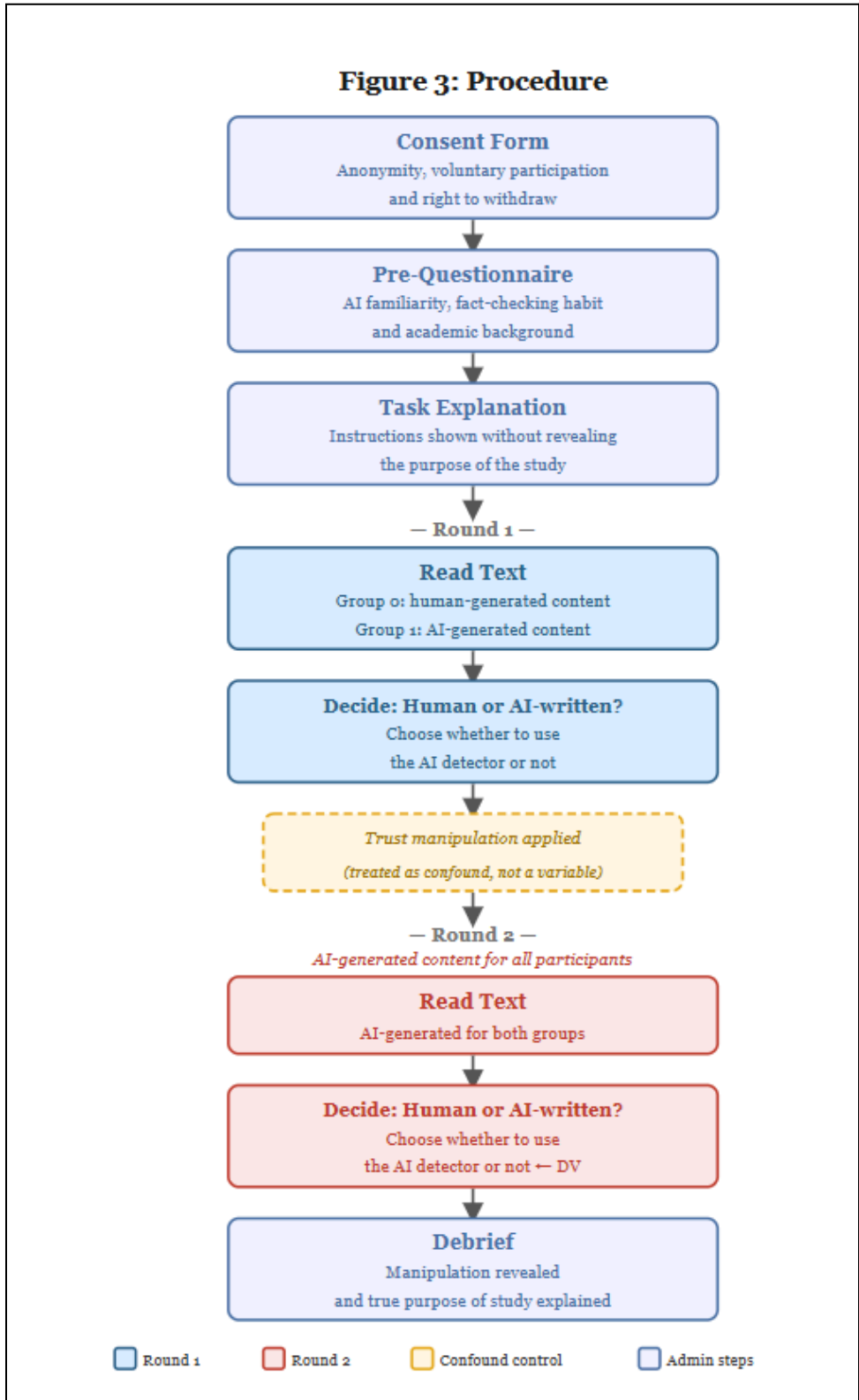




5. Participants & Sampling

We aim to recruit at least 50 participants with a minimum of 25 per group. The age range is 18 to 30, reflecting a generation that has grown up alongside digital content and is now increasingly exposed to AI-generated material in everyday online environments making this group both practically convenient and theoretically relevant to the research question. Participants need at least basic familiarity with AI tools, defined as a score of 2 or above on the AI familiarity scale, to ensure they can meaningfully engage with the task. Anyone currently enrolled in advanced courses in psychology, cognitive science, or AI is excluded to avoid prior knowledge of experimental methods or AI detection influencing their responses. Recruitment will take place through university mailing lists, student forums, and social media platforms, with the study running fully online via Pavlovia and taking approximately 10 to 15 minutes to complete.

6. Procedure

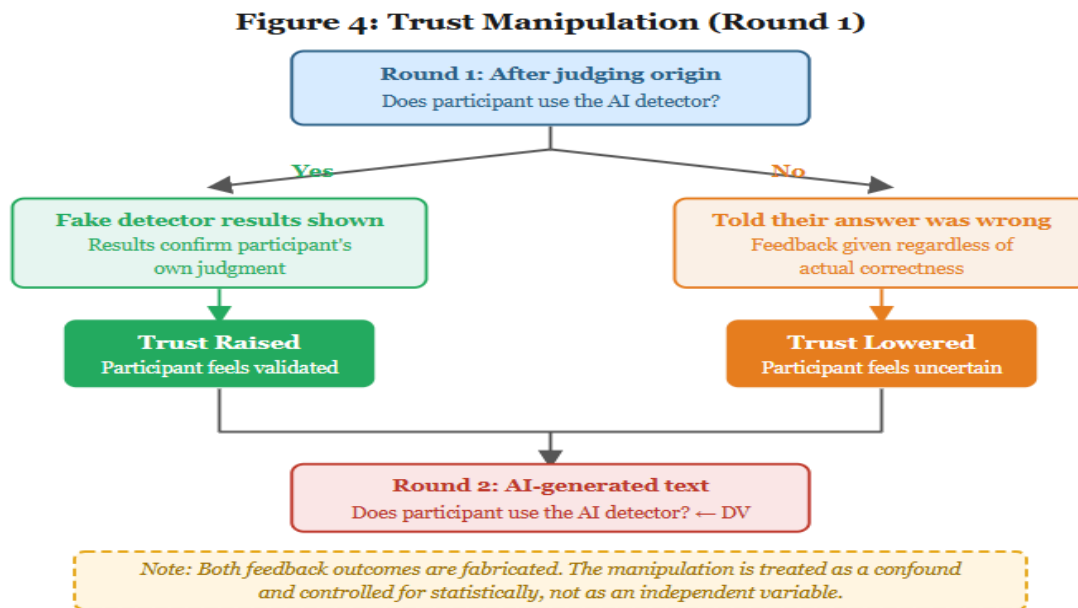


Participants first encounter an online consent form outlining the study's duration, anonymity, and their right to withdraw at any point by closing the browser window. Only participants who confirm consent are allowed to proceed.

Following consent, participants complete a short pre-questionnaire. They are asked how frequently they use AI tools in their daily life, rated on a five-point scale, and how critically they engage with content they read online, rated on a four-point scale. These responses are stored anonymously under a randomly generated participant code.

Participants then receive a brief task explanation informing them that they will read a series of short texts and answer questions about each one. The explanation deliberately omits any reference to AI detection or the purpose of the manipulation, to avoid priming participants toward a particular response.

In Round 1, participants are shown a text whose origin depends on their group assignment. Group 0 receives a human-written text and Group 1 receives an AI-generated text. After reading, participants are asked to judge whether the text was written by a human or by an AI. They are then asked whether they would like to use an AI detector to verify their judgment. This choice triggers the trust manipulation: participants who choose to use the detector are shown fabricated results from multiple AI detectors, each displaying a confidence score that confirms their own judgment, thereby raising their confidence in their detection ability. Participants who choose not to use the detector are told that their judgment was incorrect, regardless of whether it actually was, thereby lowering their confidence. This manipulation is treated as a confound and controlled statistically rather than as a variable of interest.



Round 2 is identical for all participants regardless of group. Everyone receives an AI-generated text, reads it, makes a judgment about its origin, and then decides whether to use the AI

detector. This final detector choice is the dependent variable of the study and is recorded as a binary yes or no response.

Immediately following Round 2, participants are shown a debrief screen. The debrief explains that the AI detector results shown in Round 1 were not produced by real detectors, that the confidence scores were fabricated, and that the purpose of the study was to examine how prior exposure to AI-generated content influences subsequent verification behaviour. Participants are thanked and the session ends.

7. Validity Threats & Control Measures

One potential threat to internal validity is the participant-driven nature of the trust manipulation. Whether a participant's self-trust is raised or lowered in Round 1 depends on their own choice to use the detector or not, rather than on group assignment. This means two participants in the same group can enter Round 2 with substantially different confidence levels, introducing noise that is independent of content type. We address this by recording the direction of trust change for every participant and controlling for it statistically.

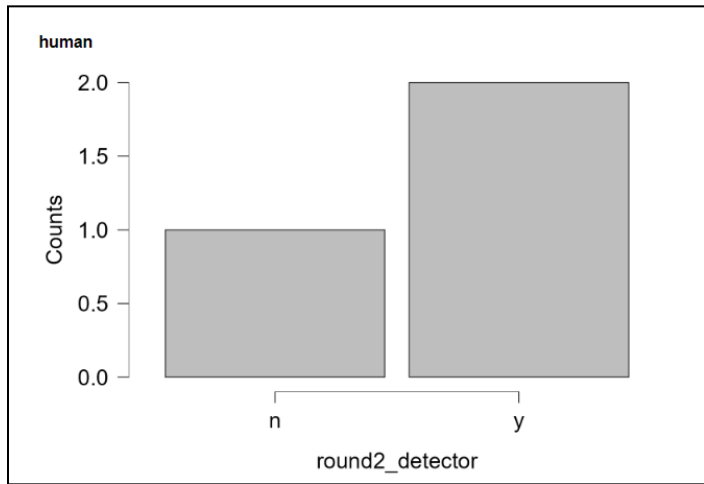
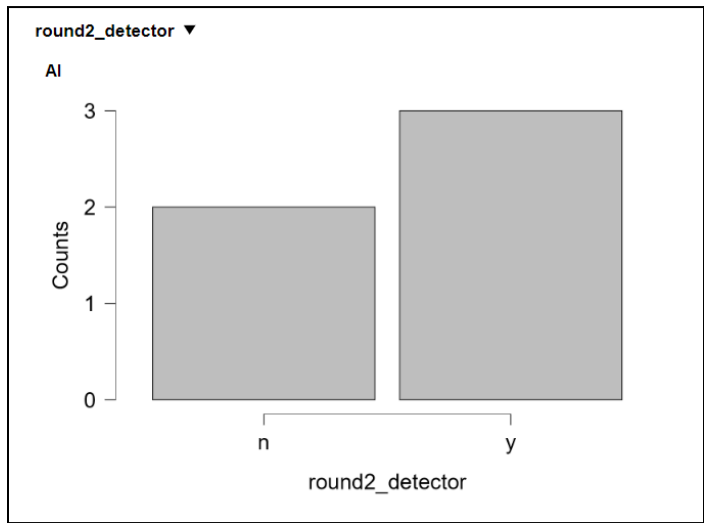
A second internal validity threat concerns the texts used in Round 1. Both the human-written and AI-generated texts are answers to the same question about the value of an interdisciplinary approach in the Human and AI master programme ensuring they are matched on topic and context. However, they may still differ in subtle qualities such as writing style, fluency, or length, any of which could independently influence the participant's Round 2 behaviour regardless of content type. This is difficult to fully control for, and represents an acknowledged limitation of the current design.

A third threat is whether participants genuinely believed the fabricated detector results shown during the manipulation. If participants were skeptical of the fake confidence scores, the trust manipulation may not have had its intended effect, weakening the confound control. We did not include a manipulation check in the current version of the study, and therefore cannot confirm believability. Future iterations should address this by asking participants after Round 1 how convincing they found the detector results.

Regarding external validity, the sample is drawn exclusively from UTN students enrolled in the Human and AI master programme. This group is likely more tech-savvy and more familiar with AI tools than the general population, which limits how broadly the findings can be generalised. Additionally, the single-exposure design means the study can only speak to the immediate effect of one encounter with AI-generated content — the theoretical argument about familiarity building over time through repeated exposure cannot be directly tested here and remains a direction for future research.

8. Pilot Data

participant	group	ai_familiarity	reading_behaviour	round1_detector	trust_condition	round2_detector
19271	human	5	3	n	lowered	n
352442	AI	5	4	y	raised	y
460830	AI	4	3	y	raised	y
518446	AI	3	3	y	raised	n
578726	human	5	4	y	raised	y
651784	AI	4	1	y	raised	y
692162	human	4	2	n	lowered	y
946073	AI	5	2	n	lowered	n



Frequency Table					
group	round2_ detector	Frequency	Percent	Valid Percent	Cumulative Percent
AI - First	n	2	40.0	40.0	40.0
	y	3	60.0	60.0	100.0
	Missing	0	0.0		
	Total	5	100.0		
Human - First	n	1	33.3	33.3	33.3
	y	2	66.7	66.7	100.0
	Missing	0	0.0		
	Total	3	100.0		

A small pilot was conducted with 8 participants, 5 assigned to the AI group and 3 to the human group. The dependent variable whether participants chose to use the AI detector in Round 2 was recorded as a binary yes/no response.

In the AI group, 3 out of 5 participants used the detector in Round 2, while 2 did not. In the human group, 2 out of 3 participants used the detector, while 1 did not. The distributions are shown in the figures above.

Contrary to the hypothesis, the AI group did not show a lower tendency to use the detector in Round 2. However, given the extremely small sample size, no statistical conclusions can be drawn from these results. The pattern could easily be due to chance with $n=8$.

9. Changes Upon Feedback

Following the professor's feedback, three main changes were made to the design.

- 1) The trust manipulation was reframed from a second independent variable to a confound controlled for statistically. This was done to sharpen the research question and avoid the interpretive ambiguity the professor flagged around mixing multiple constructs at once.
- 2) A pre-questionnaire was added to measure AI tool familiarity and online reading behaviour, providing a mechanism to control for individual differences that could independently influence the dependent variable.
- 3) Finally, the dependent variable was clarified. The original proposal mixed trust, suspicion, and detection ability without distinguishing between them. The current proposal focuses exclusively on verification behaviour — the binary detector choice in Round 2 — as the single measurable outcome.
- 4) pilot data was collected and added following the requirement to include a pilot report

10. Link to experiment

<https://run.pavlovia.org/riyazmullaji/deeptrust>

11. Data

Original Data -

<https://docs.google.com/spreadsheets/d/1oBQ9rd6FiElbv467qBX2E2Ysx7KpZQEvE9f-ZJbWHTs/edit?usp=sharing>

Cleaned Data -

<https://docs.google.com/spreadsheets/d/1JhHI3elu1-5Yu9ANMUs0NofMw5HBHLxQxzcNuajZ9jU/edit?usp=sharing>

References

Lee, H.-P., Sarkar, A., Tankelevitch, L., Drosos, I., Rintel, S., Banks, R., & Wilson, N. (2025). The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM.
<https://doi.org/10.1145/3706598.3713778>

Milička, J., Marklová, A., Drobil, O., & Pospíšilová, E. (2025). Learning to detect AI texts and learning the limits. *PLoS ONE*, 20(10), e0333007. <https://doi.org/10.1371/journal.pone.0333007>